

ER IMCQ 2

KEY

ANATOMY

KEY

A 22-year-old man comes to the emergency department because of painful swelling of the left hemiscrotum that occurred during sexual intercourse. The pain had sudden onset and was located at the left inguinal region. Physical examination shows swelling starting from the inguinal canal and extending through the entire left scrotum which was tender on palpation. A soft mass was palpable around the left testis and the spermatic cord. Laboratory tests were within normal range. Doppler ultrasonography of the scrotum shows a hematoma extending from the left superficial inguinal ring to the left hemiscrotum, partially surrounding the left testis. Both testes had normal structure and arterial blood flow. Which of the following structures is most like damaged to result in the patient's symptoms?

- A. Ductus deferens
- B. Seminiferous tubules
- C. Cremaster muscle
- D. Pampiniform plexus
- E. Testicular artery
- F. Testicular vein

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Item ID: SOM.MKII.BPM2.1.ER.1.ANAT.1047

Item Description: Describe the blood supply, venous and lymphatic drainage of the prostate gland, ductus deferens, seminal vesicles, bulbourethral glands, testes, scrotum, penis

A 35-year-old man comes to the physician because of a 2-year history of infertility despite regular unprotected intercourse with his partner. He has normal libido, spontaneous erections, and no history of testicular trauma, infections, or surgeries. Pulse is 80/min, respirations are 14/min, and blood pressure is 118/76 mm Hg. Physical examination shows normal external genitalia with bilaterally descended testes and palpable vas deferens. Testicular biopsy shows normal spermatogenesis, and transrectal ultrasound shows dilated seminal vesicles and midline cysts within the central zone of the prostate. Which of the following findings is most likely to be observed on laboratory evaluation of blood or semen?

- A. Normal sperm delivery but low semen volume
- B. Low semen volume with very low sperm count
- C. Low sperm count but normal fructose concentration
- D. Azoospermia with reduced LH and FSH
- E. Impaired testosterone production

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Item ID: SOM.MKII.BPM2.1.ER.1.ANAT.1112

Item Description: Describe the role of the seminal vesicles in the production of seminal fluid and their connection to the ejaculatory ducts.

A 64-year-old man comes to the physician for a follow-up visit 6 weeks after undergoing prostatectomy for prostate cancer. He is concerned about new-onset erectile dysfunction since the surgery. He had no issues with erections before the procedure. His PSA levels are now undetectable, and he has no urinary complaints. Injury to which of the following neural structures during surgery most likely accounts for this complication?

- A. Grey rami communicans (S2-S4)
- B. Sacral splanchnic nerves (T10 – L2)
- C. Superior hypogastric plexus
- D. Inferior hypogastric plexus
- E. Ilioinguinal nerve
- F. Cavernous nerves from the prostatic plexus
- G. Pelvic splanchnic nerves (S2–S4)
- H. Pudendal nerve
- I. Genitofemoral nerve

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- G. Pelvic splanchnic nerves (S2–S4)
- H. Pudendal nerve
- I. Genitofemoral nerve

Item ID: SOM.MKII.BPM2.1.ER.1.ANAT.1017

Item Description: Describe the nervous pathways for control of erection including the function of the cavernous nerves

A 26-year-old woman comes to the emergency department because of sudden-onset left lower abdominal pain and mild vaginal spotting. Her last menstrual period was 6 weeks ago. She has no significant medical history and is not using contraception. Her pulse is 96/min, respirations are 16/min, and blood pressure is 116/74 mm Hg. Physical examination shows tenderness over the left adnexa with rebound on deep palpation. Pelvic ultrasound shows a gestational sac implanted at the left uterine cornu with no fetal pole and a small amount of pelvic free fluid. The anterior aspect of this region is the anatomical site of origin for which of the following structures, and what is the primary lymphatic drainage of this structure?

- A. Ovarian ligament; internal iliac nodes
- B. Round ligament of the uterus; superficial inguinal nodes
- C. Pubocervical ligaments; internal iliac nodes
- D. Uterosacral ligament; internal iliac nodes
- E. Transverse cervical ligaments; internal iliac nodes
- F. Suspensory ligament of the ovary; lumbar nodes

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- E. Transverse cervical ligaments; internal iliac nodes
- F. Suspensory ligament of the ovary; lumbar nodes

Item ID: SOM.MKII.BPM2.1.ER.1.ANAT.1033
Item Description: Describe the blood supply, venous and lymphatic drainage of the ovaries, uterine tubes, fundus, body and cervix of uterus, pelvic part of vagina, vaginal vestibule.

A 32-year-old woman comes to the physician for a routine gynecologic examination. She has no complaints and her medical history is unremarkable. She is sexually active and uses a copper intrauterine device (IUD) for contraception. Her last menstrual period was 3 weeks ago. Vital signs are within normal limits. On bimanual pelvic examination, the cervix is palpated normally. However, the uterine fundus is difficult to palpate anteriorly and is more readily appreciated through the rectovaginal examination. Transvaginal ultrasonography confirms that the uterine body is tilted posteriorly toward the sacrum. Which of the following best describes the position of the uterus in this patient?

- A. Anteverted and anteflexed
- B. Anteverted and retroflexed
- C. Retroverted and retroflexed
- D. Retroverted and anteflexed
- E. Midposition and anteflexed

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- B. Anteverted and retroflexed
- C. Retroverted and retroflexed
- D. Retroverted and anteflexed
- E. Midposition and anteflexed

Item ID: SOM.MKII.BPM2.1.ER.1.ANAT.1024

Item Description: Define the terms anteverted, anteflexed, retroverted and retroflexed with respect to the position of the uterus.

A 22-year-old man is brought to the emergency department by paramedics after he fell onto a metal railing while skateboarding. He landed straddling the bar and immediately developed penile pain and swelling. His vital signs are within normal limits. Physical examination shows he is alert and hemodynamically stable. He has a swollen penis with significant ecchymosis extending into the scrotum and perineum as shown in the image. Which of the following nerves innervates the affected area indicated by the arrow?



- A. Superficial perineal
- B. Deep perineal
- C. Ilioinguinal
- D. Posterior femoral cutaneous
- E. Dorsal nerve of the penis
- F. Inferior rectal

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Item ID: SOM.MKII.BPM2.1.ER.1.ANAT.1014

Item Description: Describe the course of the pudendal nerve and its branches within the male perineum.

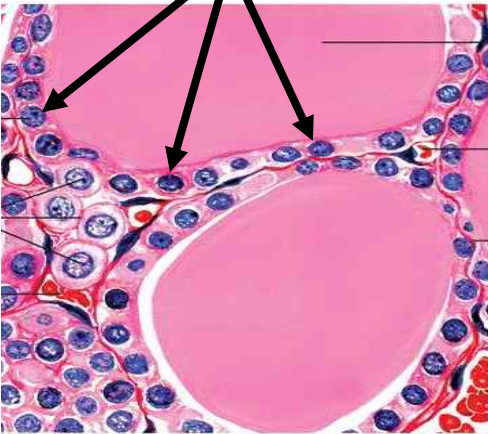
HCB Key

A 42-year-old woman comes to the physician because of a 3-month history of shortness of breath, tremors, diarrhea, palpitations, and heat intolerance. Her blood pressure is 145/70 mm Hg, pulse is 112/min, temperature is 38°C (100.4°F), and respirations are 16/min. Physical examination shows a diffusely enlarged swelling in the anterior neck, and bilateral exophthalmos. Laboratory findings are shown in the table labeled “A” below. Examination of biopsy specimens from the swelling in the neck shows hyperplasia of the cells indicated by the arrows in the normal photomicrograph labeled “B”. Which of the following best describes the regulation mechanism of the hormone secreted from the affected cell type?

- A. Inhibited by dopamine
- B. Stimulated by GHRH
- C. Stimulated by elevated calcium levels
- D. Inhibited by somatostatin
- E. Stimulated by TSH
- F. Stimulated by low calcium levels

A

	Patient's value	Normal range
TSH	0.002 mIU/L	0.5 – 5.0 mIU/L
Thyroxine	57 mcg/dL	5 – 12 mcg/dL



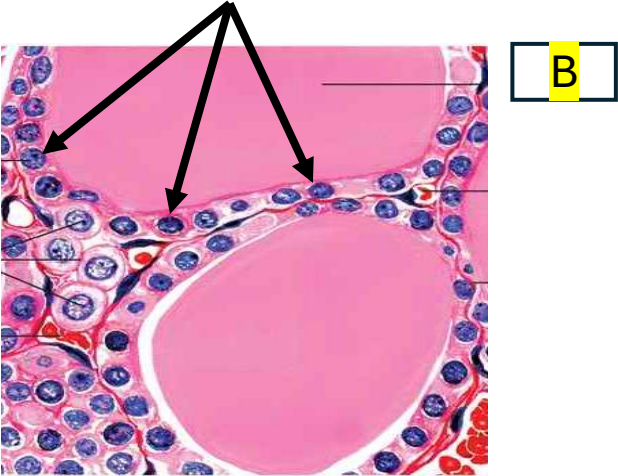
B

A 42-year-old woman comes to the physician because of a 3-month history of shortness of breath, tremors, diarrhea, palpitations, and heat intolerance. Her blood pressure is 145/70 mm Hg, pulse is 112/min, temperature is 38°C (100.4°F), and respirations are 16/min. Physical examination shows a diffusely enlarged swelling in the anterior neck, and bilateral exophthalmos. Laboratory findings are shown in the table labeled “A” below. Examination of biopsy specimens from the swelling in the neck shows hyperplasia of the cells indicated by the arrows in the normal photomicrograph labeled “B”. Which of the following best describes the regulation mechanism of the hormone secreted from the affected cell type?

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	Patient's value	Normal range
TSH	0.002 mIU/L	0.5 – 5.0 mIU/L
Thyroxine	57 mcg/dL	5 – 12 mcg/dL



A 29-year-old woman comes to the physician because of a 6-month history of progressive fatigue, weight gain of a 6.80-kg (15-lb), secondary amenorrhea, and cold intolerance. Her pulse is 50/min and respirations are 16/min. Imaging studies of her head shows atrophic changes to the pituitary gland. Which of the following blood vessels is most commonly associated with the affected gland?

- A. Elastic arteries
- B. Post capillary venules
- C. Fenestrated capillaries
- D. Muscular venules
- E. Large veins

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A 25-year-old man comes to the physician because of primary infertility. Physical examination shows reduced facial hair and testicular atrophy. Laboratory studies show decreased total testosterone and increased luteinizing hormone. A semen analysis shows azoospermia. Examination of a biopsy specimen from the testis shows the absence of the cells circumscribed in the attached normal photomicrograph. Which of the following best characterizes the cell type most likely affected that resulted in the observed laboratory and biopsy findings?

- A. Eosinophilic staining cytoplasm with numerous lipid droplets
- B. Round with sparse cytoplasm and dense heterochromatic nuclei
- C. Flat and pointed head with an axonemal complex in the tail
- D. Flattened cells found around the seminiferous tubules and contain actin filaments
- E. Columnar cells with euchromatic nucleus and prominent nucleolus



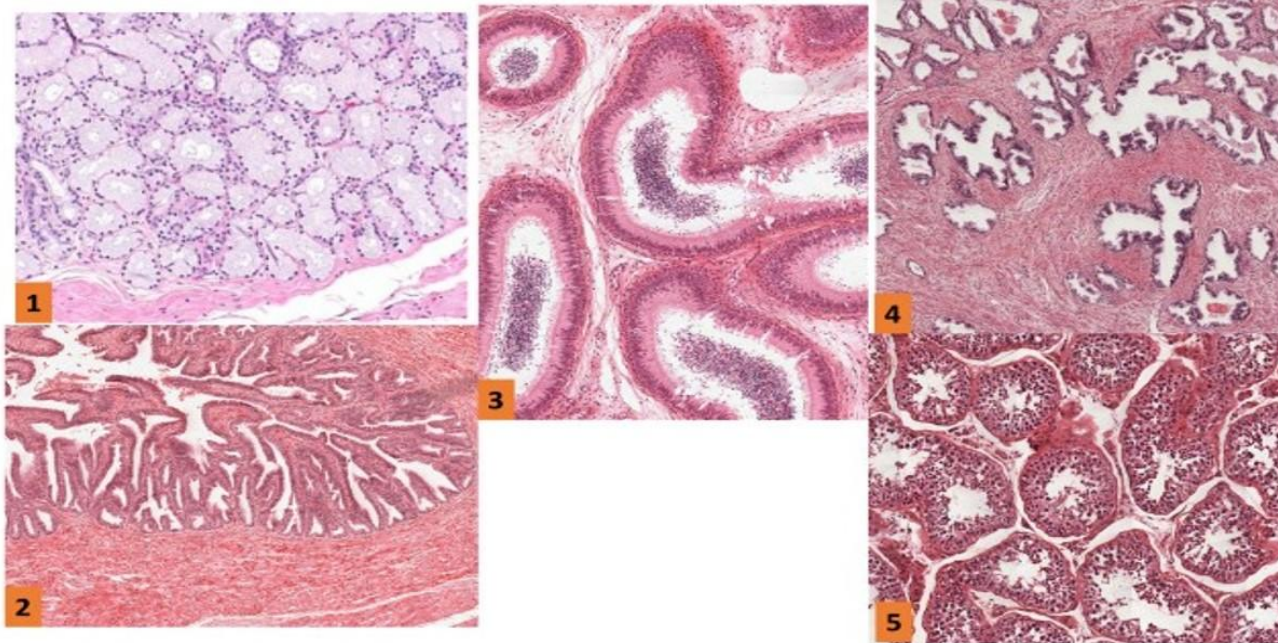
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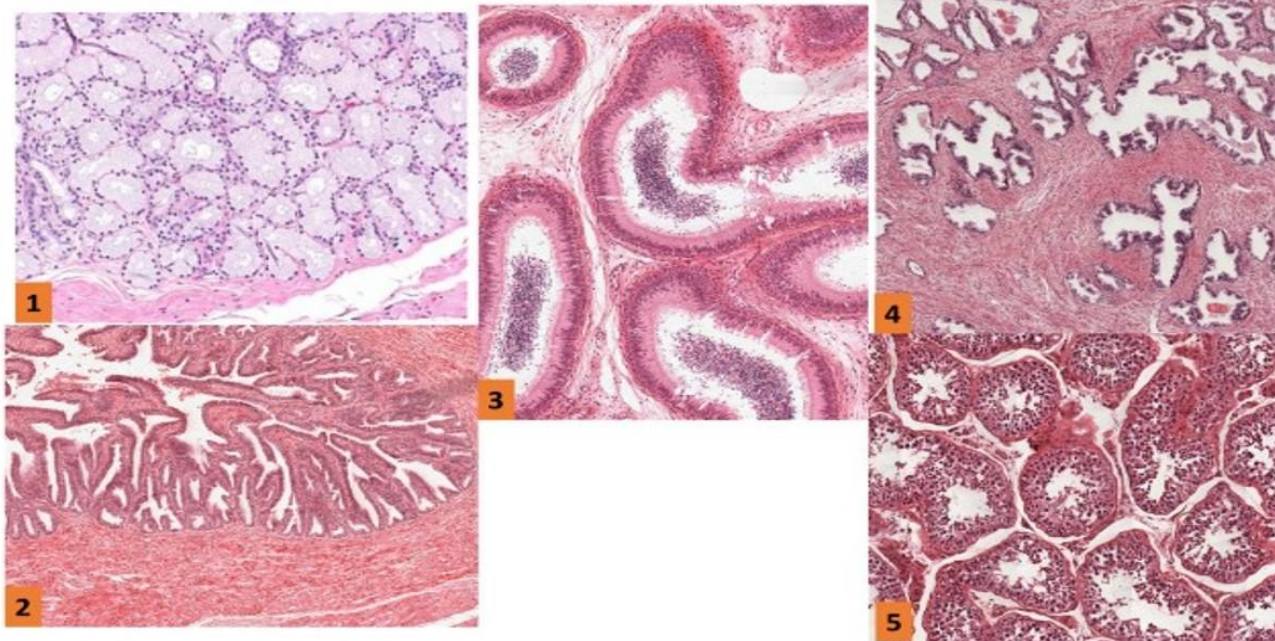
A 63-year-old man is brought to the physician because of a 2-day history of fever and chills and a 5-day history of excessive discharge from his penis that soaks through his undergarments. He underwent a transurethral resection of the prostate (TURP) 8 months ago. His temperature is 40°C (104°F) and pulse is 102/min. Physical examination shows a clear liquid at the external urethral meatus. A urine sample shows a long thread of mucus mixed in with the urine. Laboratory studies of the fluid show large amounts of mucus, galactose and sialic acid. The physician suspects an abscess of an accessory sex gland. Based on the physical examination findings and laboratory studies, which of the following photomicrographs best identifies the affected tissue?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



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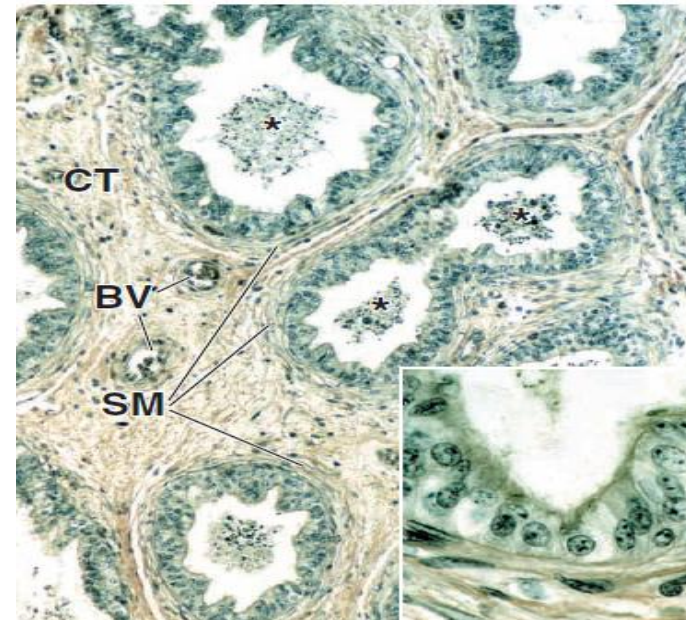
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SOM.MK.I.BPM2.2.ER.1.HCB.1107 State the function of the bulbourethral glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1108 Describe the microscopic structure of the bulbourethral glands.

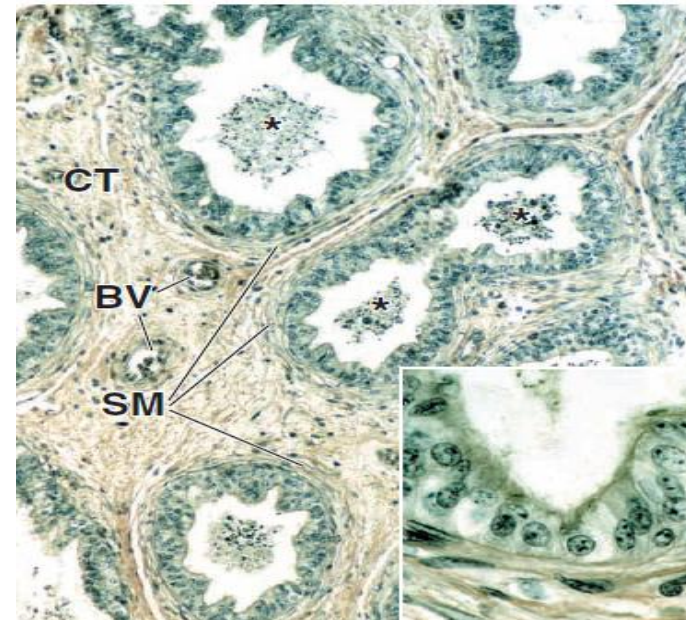
A 32-year-old man comes to the physician because of a 2-day history of bilateral scrotal pain, worse upon ejaculation. He has a 5-year history of primary infertility. Physical examination shows a mass at the head of the left epididymis. Examination of a biopsy specimen of this mass is shown in the photomicrograph. Which of the following best describes a function of the affected structure?

- A. Storage of sperm
- B. Reabsorption of excess fluid
- C. Emission of semen
- D. Production of sperm
- E. Formation of blood-testis barrier



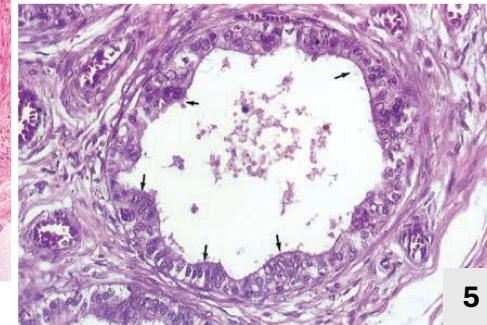
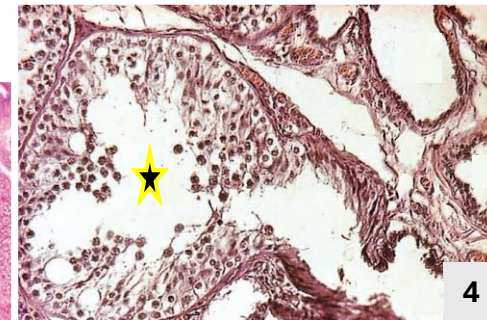
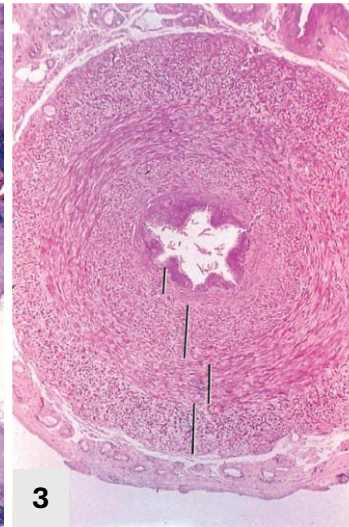
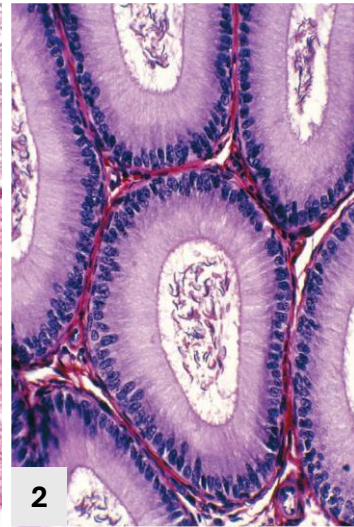
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A 14-year-old boy is brought to the emergency department by his parents because of a sudden onset of severe pain in the left testicle, while playing football. He also complains of nausea but no vomiting. Physical exam of the left scrotum shows a swollen, erythematous, hard left testicle in a high and transverse position. Ultrasonography shows a hydrocele with reduced blood flow to the left testis. Blood flow to the right testis is unaffected. He is scheduled for immediate treatment to reduce the risk of ischemia and testicular necrosis due to obstruction of the blood vessels. Which of the following best represents an additional structure most likely to be obstructed in this patient?

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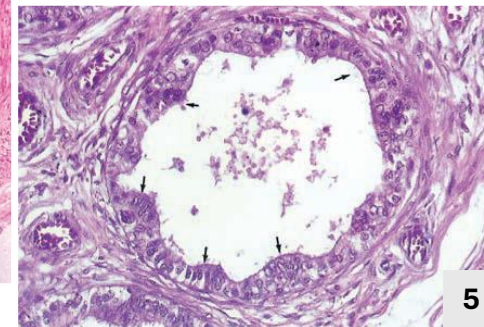
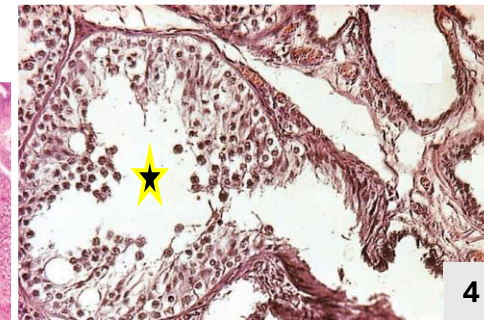
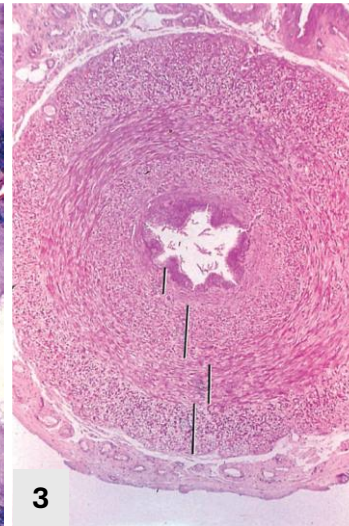
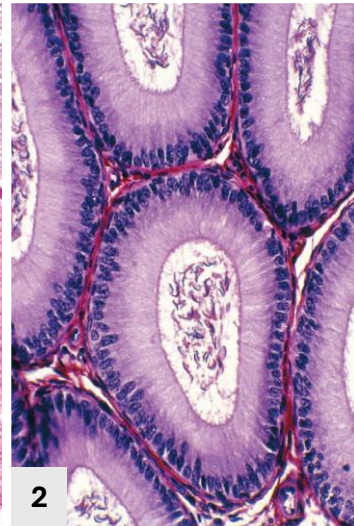
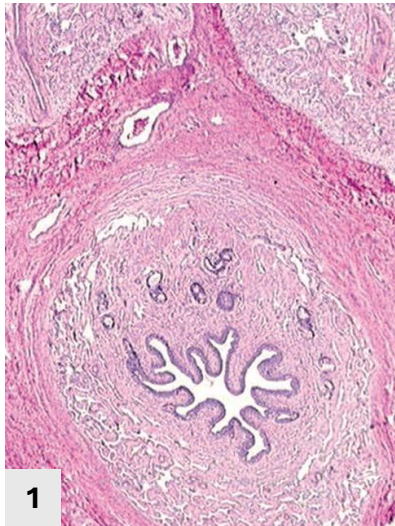
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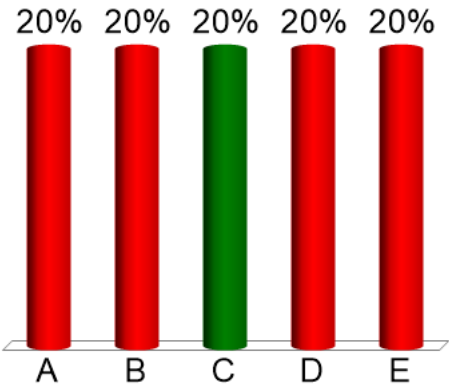


PHYSIOLOGY

KEY

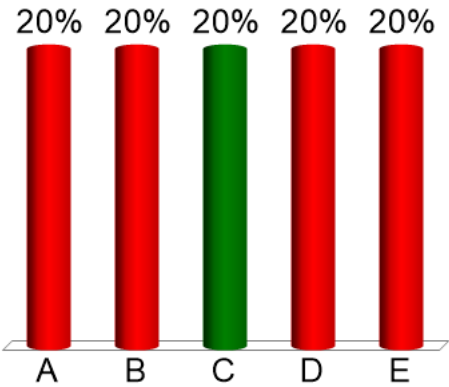
1. A 5-year-old girl is brought to the physician by her mother because of a 4-week history of recurrent muscle cramps and developmental delay since birth. Vital signs are within normal limits. Physical examination shows short stature, dental hypoplasia, short fingers, and a round face. Genetic study shows a mutation in the PTH receptor gene. She is diagnosed with pseudohypoparathyroidism. Which of the following sets of serum alterations is most likely in this patient?

Option	PTH	Calcium	Phosphate
A	No change	↓	No change
B	↓	↓	↓
C	↑	↓	↑
D	↓	↓	No change
E	↑	↑	↓



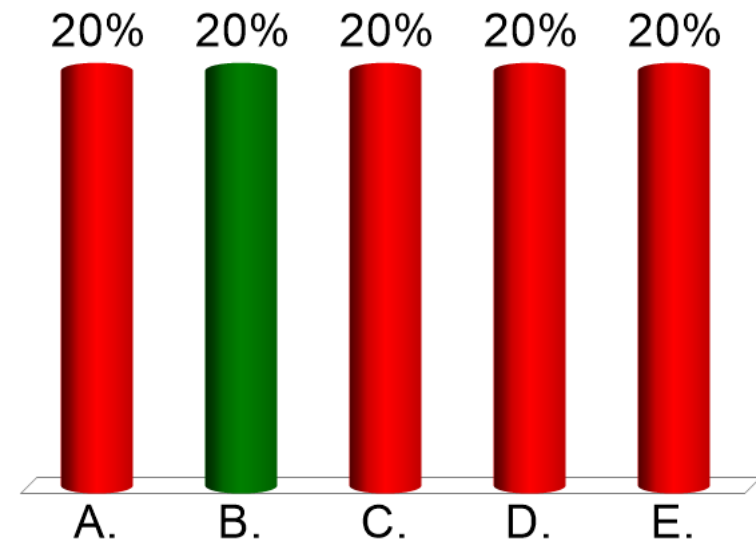
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D	↓	↓	No change
E	↑	↑	↓



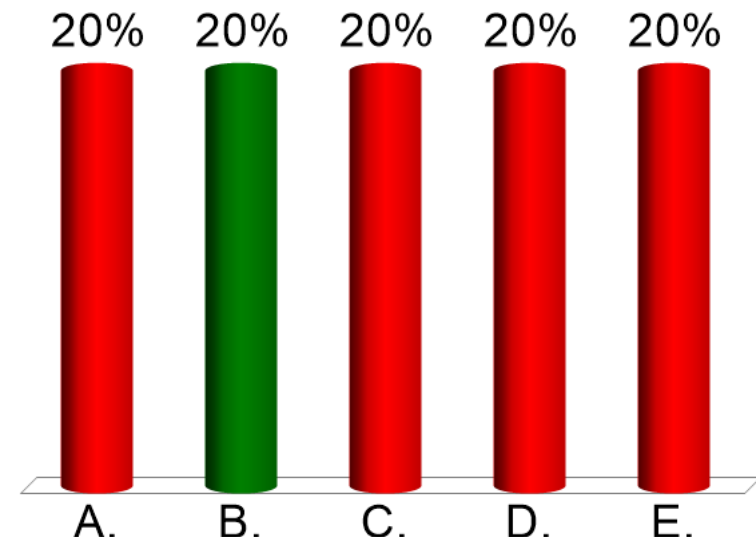
2. A 16-year-old boy is brought to the office for a well-child visit. The parents report that his teacher has expressed concerns about his reading and writing skills. Vital signs are within normal limits. Physical examination shows bilateral gynecomastia, small testes, and scanty pubic and axillary hair. Genetic studies show 47,XXY karyotype. Which of the following additional physical findings is most likely to be present in this patient?

- A. Increase muscle mass
- B. Increased linear height
- C. Loss of scalp hair
- D. Increased acne on face and shoulders
- E. Deepening of the voice



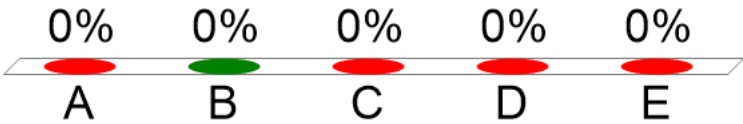
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3. A 60-year-old man comes to the physician because of a 5-week history of 3-kg (6.6-lb) weight loss and difficulty urinating. Vital signs are within normal limits. Physical examination shows normal facial and pubic hair. Digital rectal examination shows a hard nodule in the prostate. Histologic examination of a biopsy specimen of this nodule shows prostate cancer. He is treated with a drug that stimulates the secretion of gonadotropin releasing hormone continuously and not in a pulsatile manner. Which of the following sets of hormonal alterations is most likely in this patient if this treatment is continued for a long time?

Option	Follicle stimulating hormone	Luteinizing hormone	Testosterone
A	No change	↓	No change
B	↓	↓	↓
C	No change	↑	↑
D	↓	↓	No change
E	↑	↑	↓



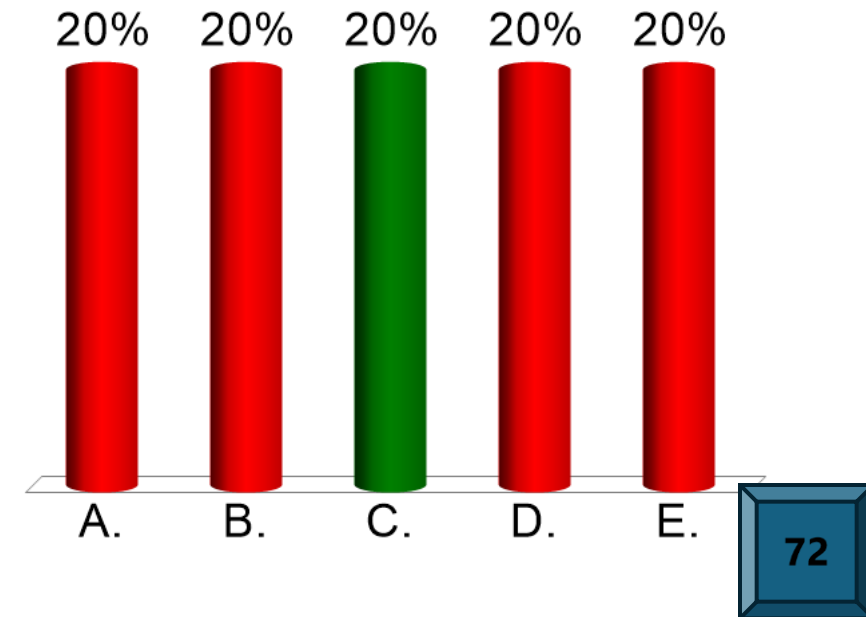
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C	No change	↑	↑
D	↓	↓	No change
E	↑	↑	↓

SOM.MK.II.BPM2.1.ER.1.PHYS. 1068. Describe each of these hormones GnRH , FSH, LH and their mechanism of action

4. A 26-year-old man comes to the clinic because of a 2-year history of infertility. Vital signs are within normal limits. Physical examination shows normal distribution of facial, axillary, body, and pubic hair. Genital examination shows small firm testes. Laboratory studies show decreased sperm count but normal plasma testosterone concentration. Which of the following is the most likely cause of these findings?

- A. Kallmann syndrome
- B. 5-alpha reductase deficiency
- C. Androgen use
- D. Hyperprolactinemia
- E. Klinefelter syndrome



4. A 26-year-old man comes to the clinic because of a 2-year history of infertility. Vital signs are within normal limits. Physical examination shows normal distribution of facial, axillary, body, and pubic hair. Genital examination shows small firm testes. Laboratory studies show decreased sperm count but normal plasma testosterone concentration. Which of the following is the most likely cause of these findings?

- A. Kallmann syndrome
- B. 5-alpha reductase deficiency
- ✓ C. Androgen use
- D. Hyperprolactinemia
- E. Klinefelter syndrome

